Range

int : $[-2\times10^9, +2\times10^9]$

long : $[-9\times10^{18}, +9\times10^{18}]$

Modular Arithmetic → +, /, -, *

$\boxed{\text{Modulo} \rightarrow \%}$

" n modulo a "

$\boxed{n \% a}$ → Remainder when n is divided by a!

$\boxed{10 \% 3 = 1}$

$\boxed{13 \% 5 = 3}$

divisor 3, 10 → Dividend, 3 → Quotient, −9, 1 → Remainder

Divisor 5, 13 → Divi., 2 → Q, −10, 3 → R

$10 = 3 \times 3 + 1$

$13 = 5 \times 2 + 3$

$\boxed{\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}}$

$\boxed{\text{remainder} = \text{Dividend} - \text{Divisor} \times \text{Quotient}}$

$32 \% 5 = 32 - 5 \times \underline{6}$

$= 32 - 30 = 2$

$17 \% 4 = 17 - 16 = 1$

$\boxed{\text{Remainder : Dividend} - \text{greatest multiple of divisor} <= \text{dividend}}$

$x \% 3$

$\boxed{0 <= \text{remainder} < \text{divisor}}$

0 1 2

$\boxed{a \% b \longrightarrow [0, b-1]}$

$\overset{\times 8}{32 \% 4} \longrightarrow 0$

$\boxed{32 = 4 \times 8 + 0}$

$12 \% 6 \longrightarrow 0$

$\times 2$

$a = K . b$ $\quad K \geq 0$

$\boxed{a \% b = 0}$

$\boxed{\text{if } ((a \% b) == 0) \longrightarrow a \text{ is a multiple of } b}$

- $150 \% 11 \rightarrow 150 -$ greatest multiple of 11 $<= 150$

  $150 - 143$

  $= 7$

- $100 \% 7 \rightarrow 100 -$ greatest multiple of 7 $<= 100$

  $100 - 7 \times 14$

  $100 - 98 = 2$

- $5 \% 12 \rightarrow 5 -$ greatest mult of 12 $<= 5$

  $5 - 12 \times 0$

  $5 - 0 = 5$

- $7 \% 15 = 7 - 15 \times 0$

  $7 - 15 \times 0 = 7$

$$\boxed{a \% b \rightarrow a : a < b}$$

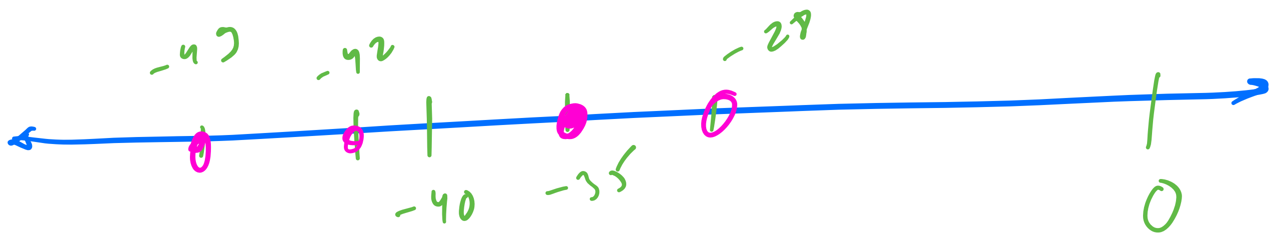
⊙ $-40 \% 7 = -40 -$ greatest mult. of $7 <= -40$

$[0, 6]$

$-40 - (-42)$

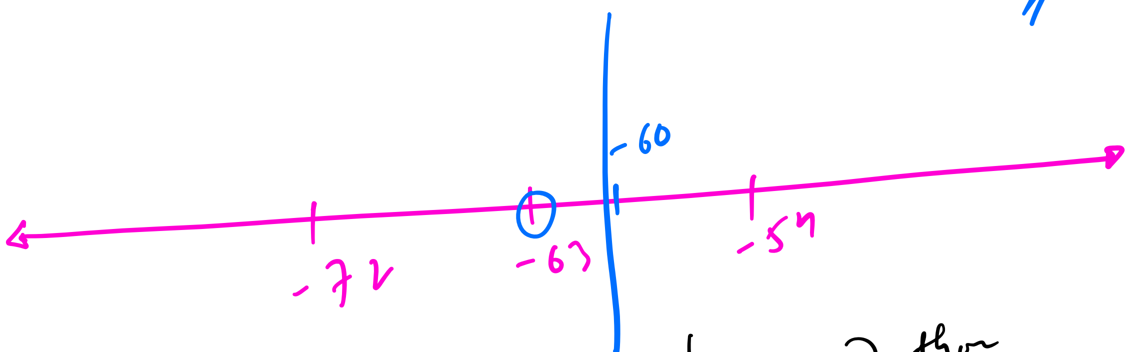
$-40 + 42 = 2$

two



⊙ $-60 \% 9 \rightarrow -60 -$ greatest mult. of $9 <= -60$

$-60 - (-63) = 3$



C++ / Java		Python
$-40 \% 7 \rightarrow -5 \xrightarrow{+7} 2$		$-40 \% 7 \rightarrow 2$
$-60 \% 9 \rightarrow -6 \xrightarrow{+9} 3$		$-60 \% 9 \rightarrow 3$

```
r = a % b;
if (r<0) r += b;
```

C++

$36 \% 7 \rightarrow 36 -$ greatest mult. of 7 $<= 36$

$7 \times x <= 36$

$x <= (36/7) \rightarrow \underbrace{5....}_{5}$

$36 - 7 \times (36/7)$

$36 - 7 \times 5$

$36 - 35 = 1$

$$\boxed{a \% b = a - (a/b) \times b}$$

$-40 \% 7 = -40 - (-40/7) \times 7$

$-40 - (-5) \times 7$

$-40 + 35 = -5 \xrightarrow{+7} 2$

python

$36 \% 7 \longrightarrow 36 - (36/7) \times 7$

$36 - 5.14\ldots \times 7 \rightarrow 0.\sim$

INSTEAD

$36 - \text{floor}(36/7) \times 7$

$36 - \text{floor}(5.\ldots) \times 7$

$36 - 5 \times 7$

$36 - 35 = 1$

$\boxed{\text{floor}(x) = \lfloor x \rfloor = \text{greatest integer} <= x}$

$\lfloor 5.7 \rfloor = 5$

$\lfloor 6 \rfloor = 6$

$$\boxed{a \% b = a - \text{floor}(a/b) \times b}$$

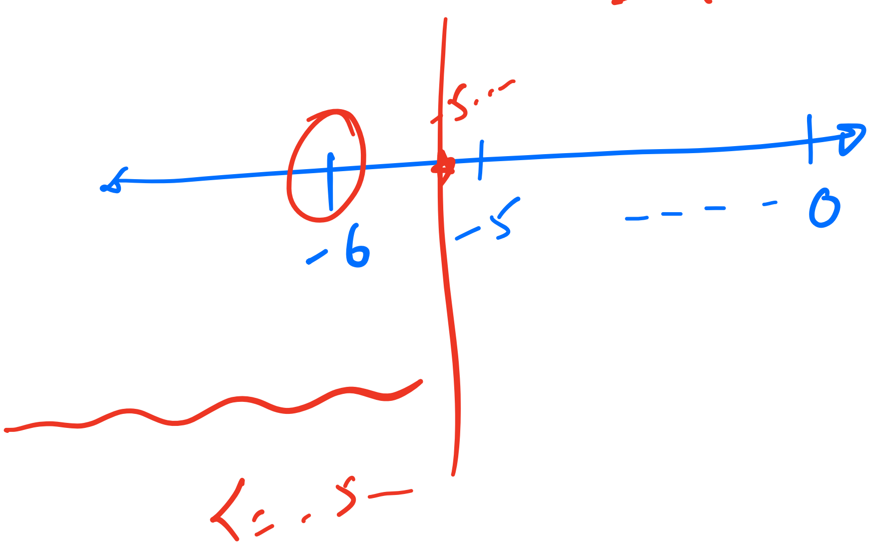
$-40 \% 7 = -40 - \text{floor}(-40/7) \times 7$

$-40 - \text{floor}(-5.\ldots) \times 7$

$-40 - (-6) \times 7$

$-40 + 42$

$= 2$



⊙ Why % ?

- Haven't used much
- HashMap / HashSet
- Circular array / Queue
- Consistent Hashing
- Cryptography
- ⋮

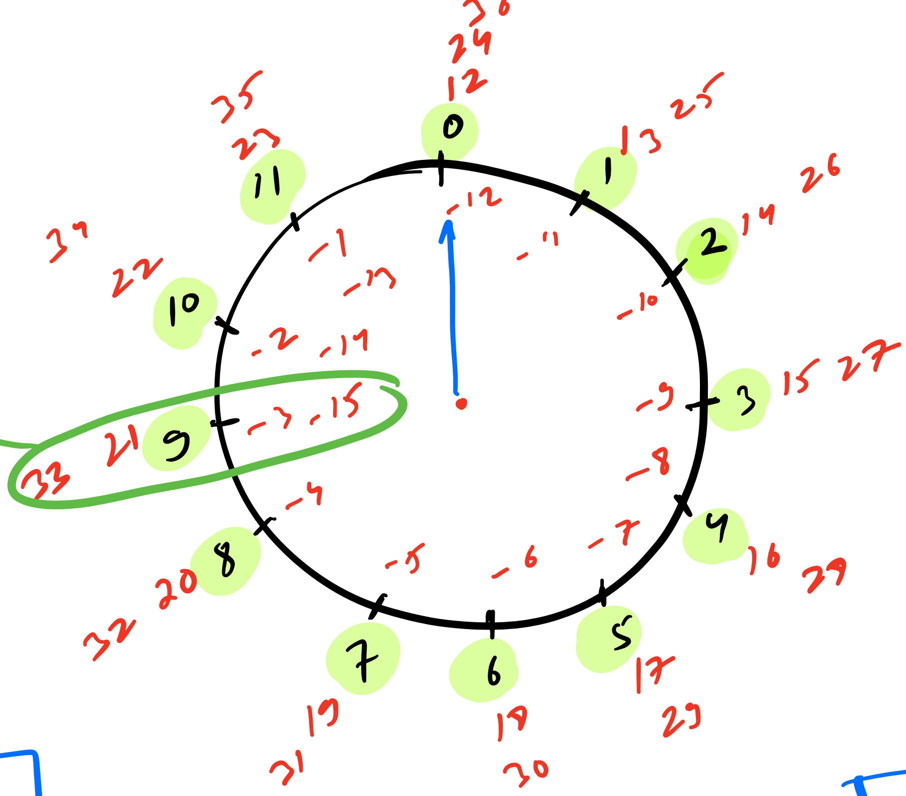
$-\infty$

$+\infty$

$\% b \rightarrow$ 0 … $b-1$

limit the data to req'd range!

0



modular congruent

$\boxed{\begin{array}{l} 8 \% 12 = 8 \\ 32 \% 12 = 8 \\ 20 \% 12 = 8 \end{array}}$

2

a, b

$\boxed{\begin{array}{l} 6 \% 12 \to 6 \\ 18 \% 12 \to 6 \\ 30 \% 12 \to 6 \end{array}}$

$\boxed{a \% n = b \% n}$ → a & b are modular congruent w.r.t n

$a = 56$

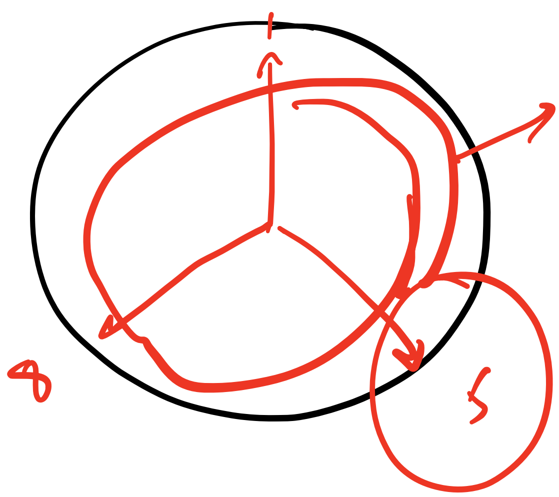
$b = 57$

$\boxed{113} \% 12 = 5$

$M = 12$

$56 \% 12 \to 8$

$57 \% 12 \to 9$



$a \% M$?

$b \% M$

④ Modular Arithmetic

1) ADDITION

$$\boxed{\underbrace{(a+b)\,\%\,M}_{[0,\,M-1]} = (\underbrace{a\,\%\,M}_{[0,\,M-1]} + \underbrace{b\,\%\,M}_{[0,\,M-1]})\,\%\,M}$$

$[0, M-1] + [0, M-1]$

↓

$[0, 2M-2] \,\%\, M$

↓

$[0, M-1]$

2) MULTIPLICATION

$$\boxed{(a \times b)\,\%\,M = ((a\,\%\,M) \times (b\,\%\,M))\,\%\,M}$$

$a = 7$

$b = 6$

$M = 4$

$(7 \times 6)\,\%\,4$

↓

$42\,\%\,4$

↓

2

$((7\,\%\,4) \times (6\,\%\,4))\,\%\,4$

↓

$(3 \times 2)\,\%\,4$

$6\,\%\,4$

↓

2

$$\boxed{a \% b = (a \% b) \% b = ((a \% b) \% b) \% b \ldots}$$

Q Given an array. find the product of all no.'s $\%\ 10^9+7$

$$\boxed{\begin{array}{l} 1 <= N <= 10^5 \\ 1 <= A[i] <= 10^9 \end{array}}$$

$$\overbrace{10^9 \times 10^9 \times 10^9 \times \ldots \times 10^9}^{10^5}$$

$10^{9 \times 10^5}$

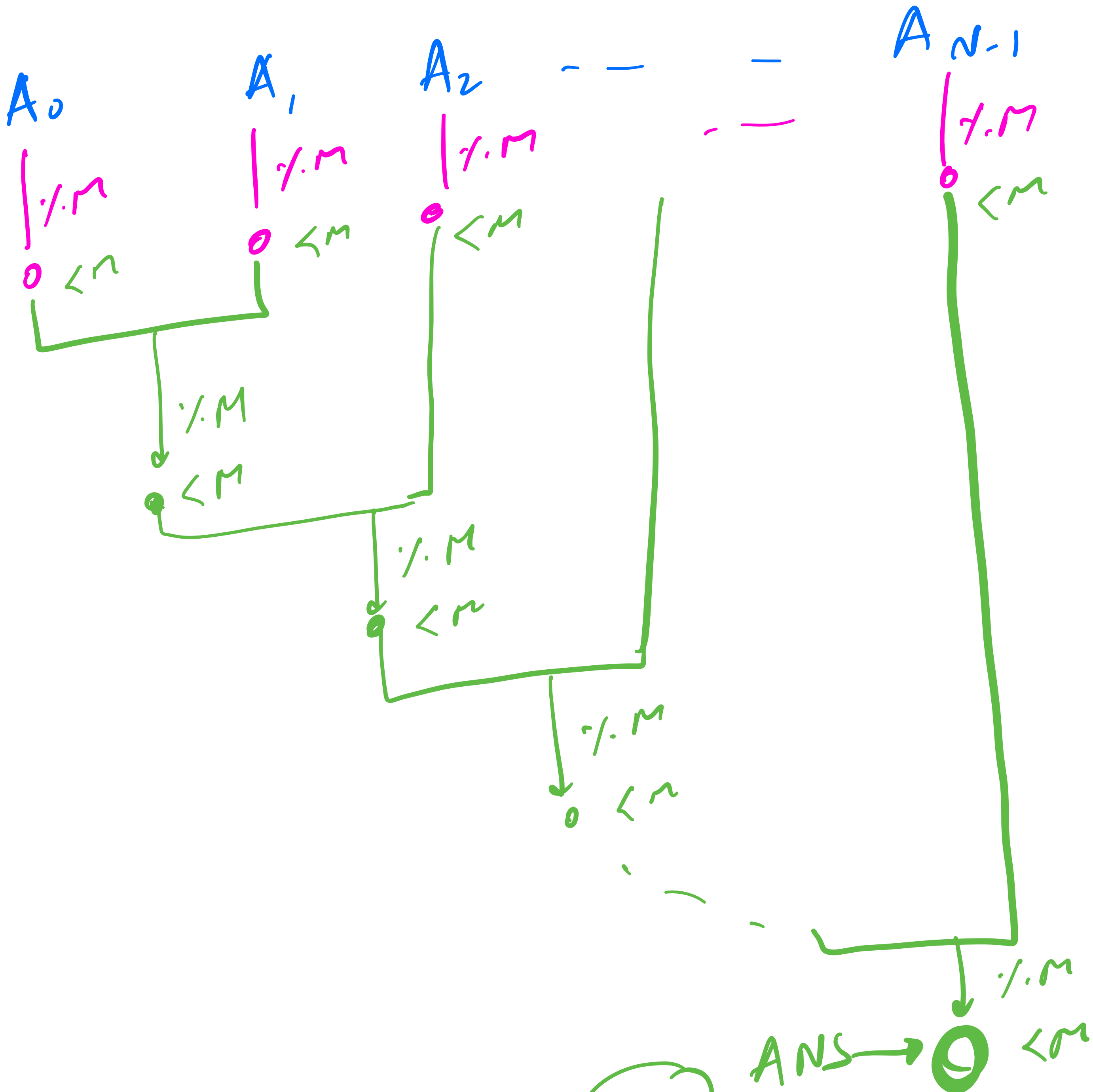
$\%\ 10^9+7$

$$\boxed{M = 10^9+7}$$

$$(\underbrace{A_0}_{x} \times \underbrace{A_1 \times A_2 \times \text{———} \times A_{N-1})}_{y} \% M$$

$$((A_0 \% M) \times (\underbrace{A_1}_{x} \times \underbrace{A_2 \times \text{——} \times A_{N-1}}_{y}) \% M) \% M$$

$$((A_0 \% M) \times (A_1 \% M) \times (A_2 \% M) \times \text{-- --} \times (A_{N-1} \% M)) \% M$$



~~int~~ long ans = 1;

f(i = 0; i < N; i++) {

ans = ($\underbrace{(ans \cancel{\% M})}_{<M}$ × $\underbrace{(A[i] \cancel{\% M})}_{<M}$) % M;

}

ret ans;

~10^9

10^9

(<M % M) → <M

~10^9 L × ~10^9 i → 10^{18} L → <M ~10^9

A[i] <= 10^9

M = ~10^9 + 7

$\boxed{TC = O(N)}$

ANS = (b × a + c) × d ;

(((((b % M) × (a % M)) % M) % M + (c % M)) % M % M × d % M) % M

HW

a^b % M

a × a × a - — × a

1 <= a <= 10^9

1 <= b <= 10^6

M = 10^9 + 7

Q Given a no. in an Array!

Calculate the no % M! → 10^9+7

$0 <= A[i] < 10$

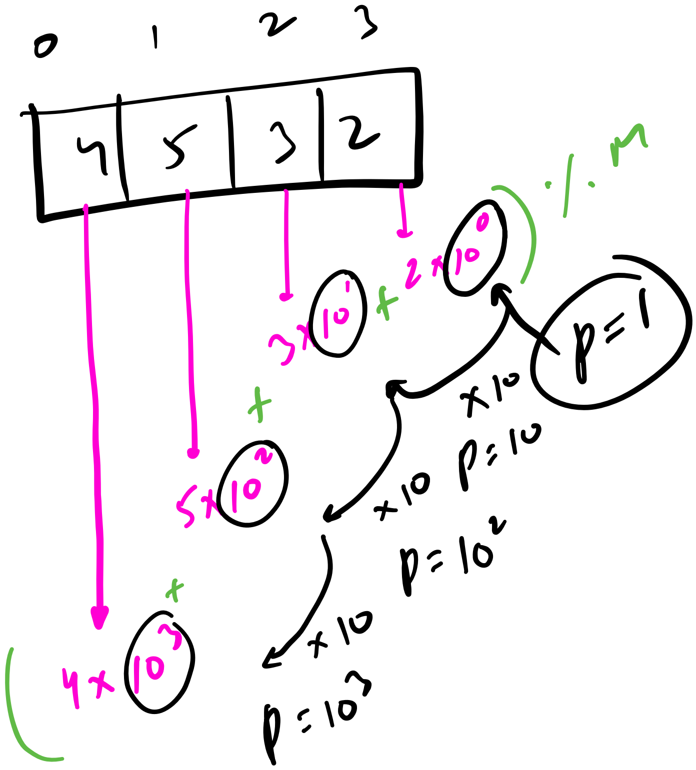
$1 <= N <= 10^5$

A :

4	5	2	6	7	8	3	2	5	3	2	5	8	4	3

$(4\,5\,2\,6\,7\,8\,3\,2\,5\,3\,2\,9\,8\,4\,3) \% 10^9+7$

?



→ int

int $p = 1$, $num = 0$;

long

f ($i = N-1$; $i >= 0$; $i--$) {

$num = ((num \cancel{\% M}) + ((A[i] \cancel{\% M}) \times (p \cancel{\% M})) \% M) \% M$

i

$< M$

$p = ((p \cancel{\% M}) \times (10 \cancel{\% M})) \% M$;

$< M$

}

ret num;

$A[i] < 10$

$M = 10^9 + 7$

II

4 5 3 6

4

$(4 \times 10) + 5$

$45 \times 10 + 3$

$453 \times 10 + 6$

HW

$num = 0$;

f($i = 0 \rightarrow N-1$) {

$num = (num \times 10) + A[i]$;

}

ret num;

Divisibility Rule

⊙ Div. Rule by 3 ?

→ Sum of digits is div. by 3

$987 \rightarrow (9+8+7)\ \%\ 3 \rightarrow 0$ ✓

$(1234)\ \%\ 3$

$= (1\times10^3 + 2\times10^2 + 3\times10^1 + 4\times10^0)\ \%\ 3$

$((1\times10^3)\%3 + (2\times10^2)\%3 + (3\times10^1)\%3 + (4\times10^0)\%3)\%3$

$(1\times(10^3)\%3 + 2\times10^2\%3 + 3\times10^1\%3 + 4\times10^0\%3)\%3$

$(1\times1 + 2\times1 + 3\times1 + 4\times1)\%3$

$(1+2+3+4)\%3$

⊙ Div. Rule for 4

→ No. formed by last 2 digits should be divisible by 4.

eg: 4856 ✓ $56 \% 4 = 0$ ✓

$(1234) \% 4$

$= (1 \times 10^3 \% 4 + 2 \times 10^2 \% 4 + 34) \% 4$

0 0 0

$(34) \% 4$